

Benchmark #1

2MeV Fe on Fe

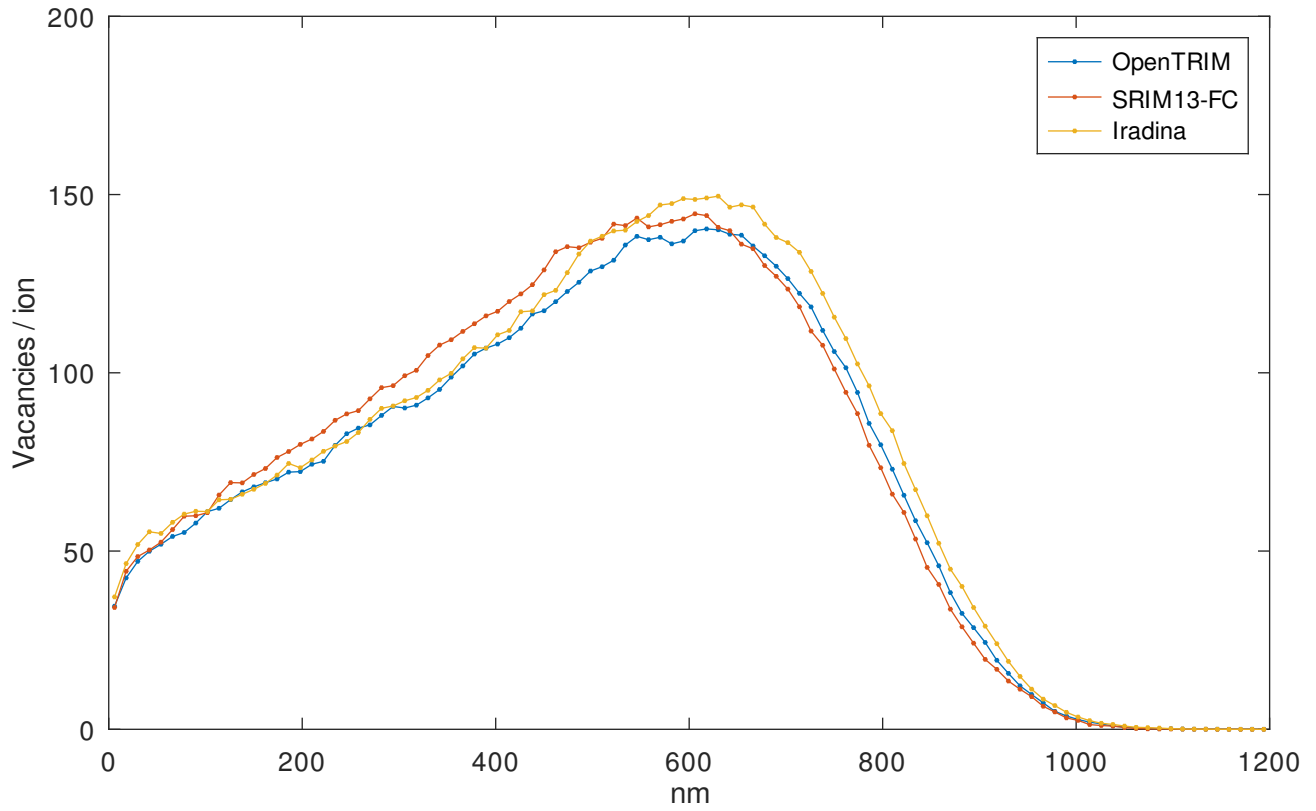
Ion energy $E_0 = 2\text{e}+06$ eV

Target depth = 1200 nm

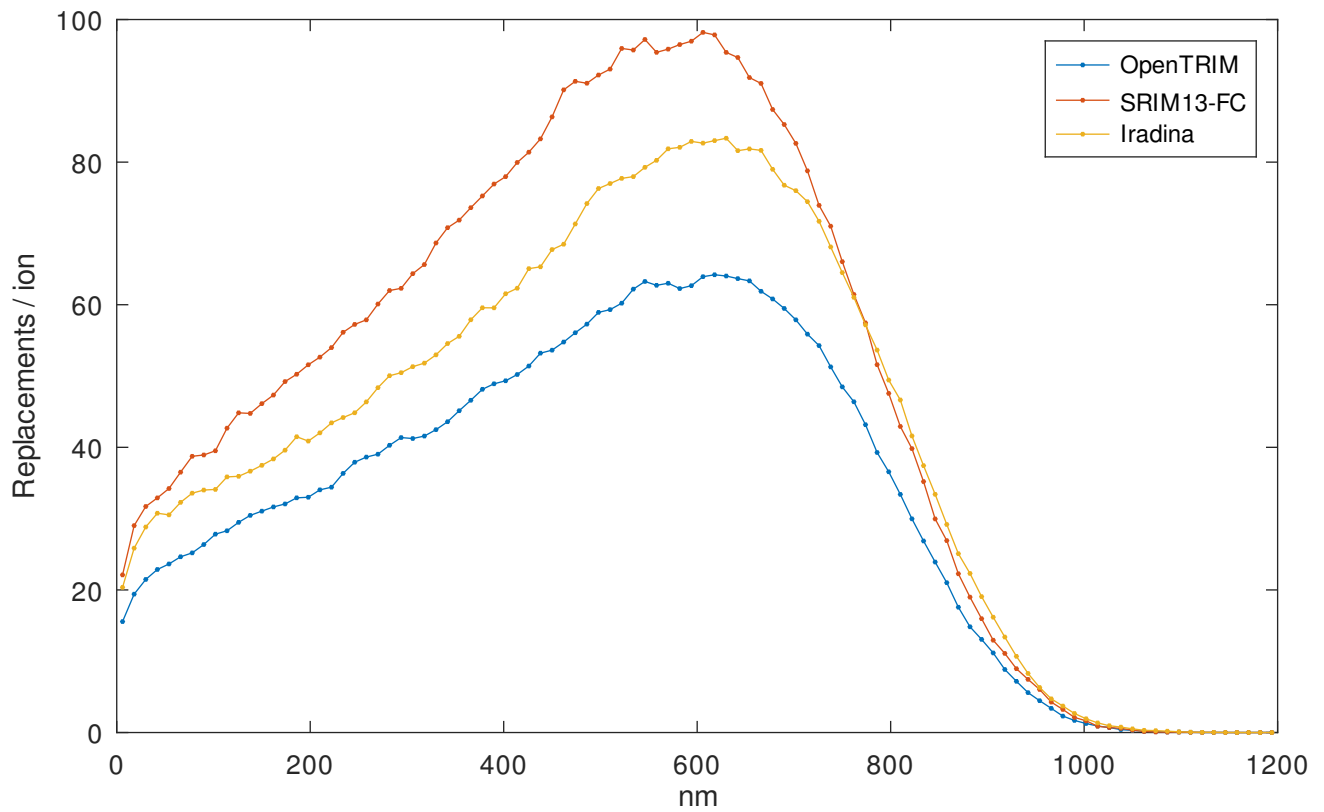
Summary Table

Quantity	OpenTRIM	SRIM13-FC	Iradina
$V(\text{Fe})$	7.13e+03	7.35e+03	7.53e+03
$R(\text{tot})$	3.26e+03	4.87e+03	4.19e+03
$I(\text{Fe})$	1	1	1
$\Delta E_e(\text{Fe})/E_0$	0.557	0.552	0
$\Delta E_e(r)/E_0$	0.112	0.112	0
$\Delta E_e(\text{tot})/E_0$	0.669	0.663	0.681
$\Delta E_n(\text{Fe})/E_0$	0.0037	0.0034	0
$\Delta E_n(r)/E_0$	0.327	0.325	0
$\Delta E_n(\text{tot})/E_0$	0.33	0.329	0.319
$1 - (\Delta E_e + \Delta E_n)/E_0$	0.000285	0.00772	0.000285

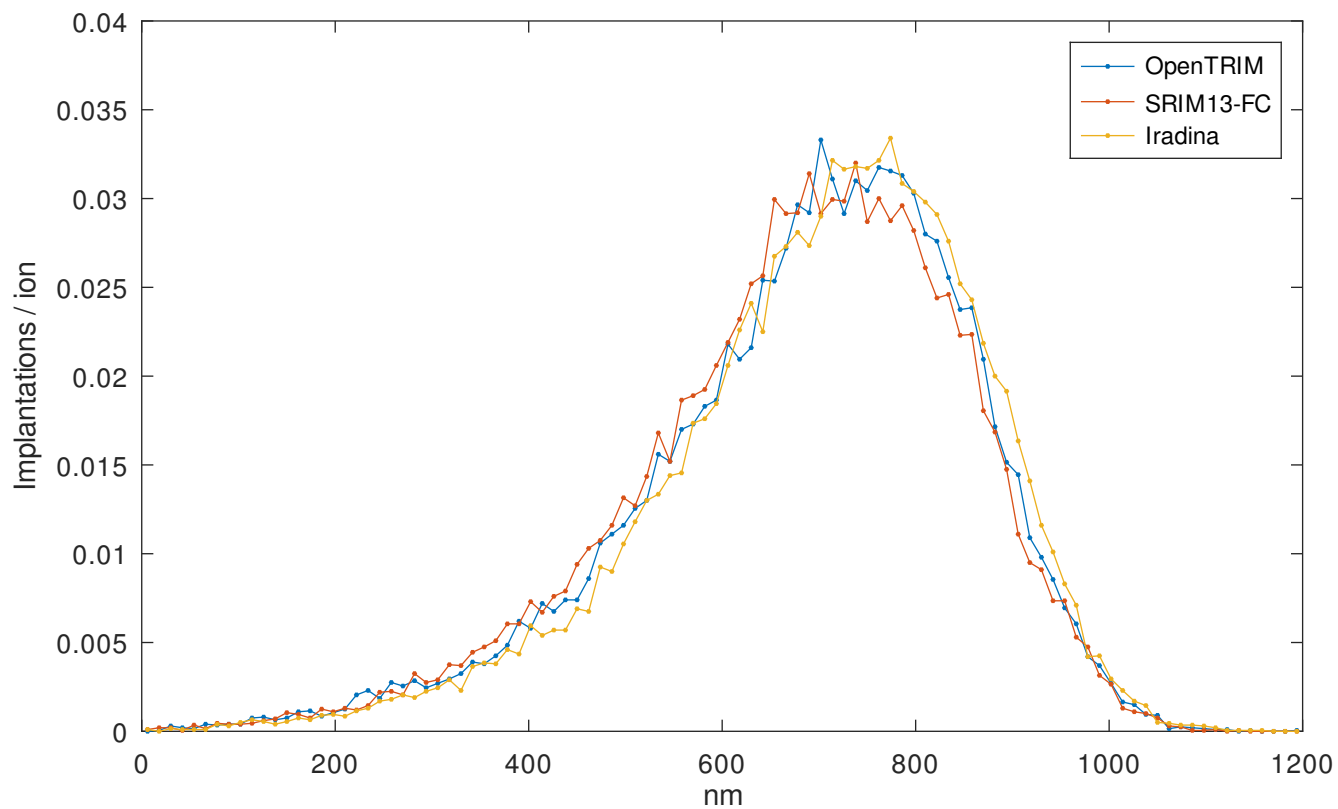
Vacancies of Fe in Fe



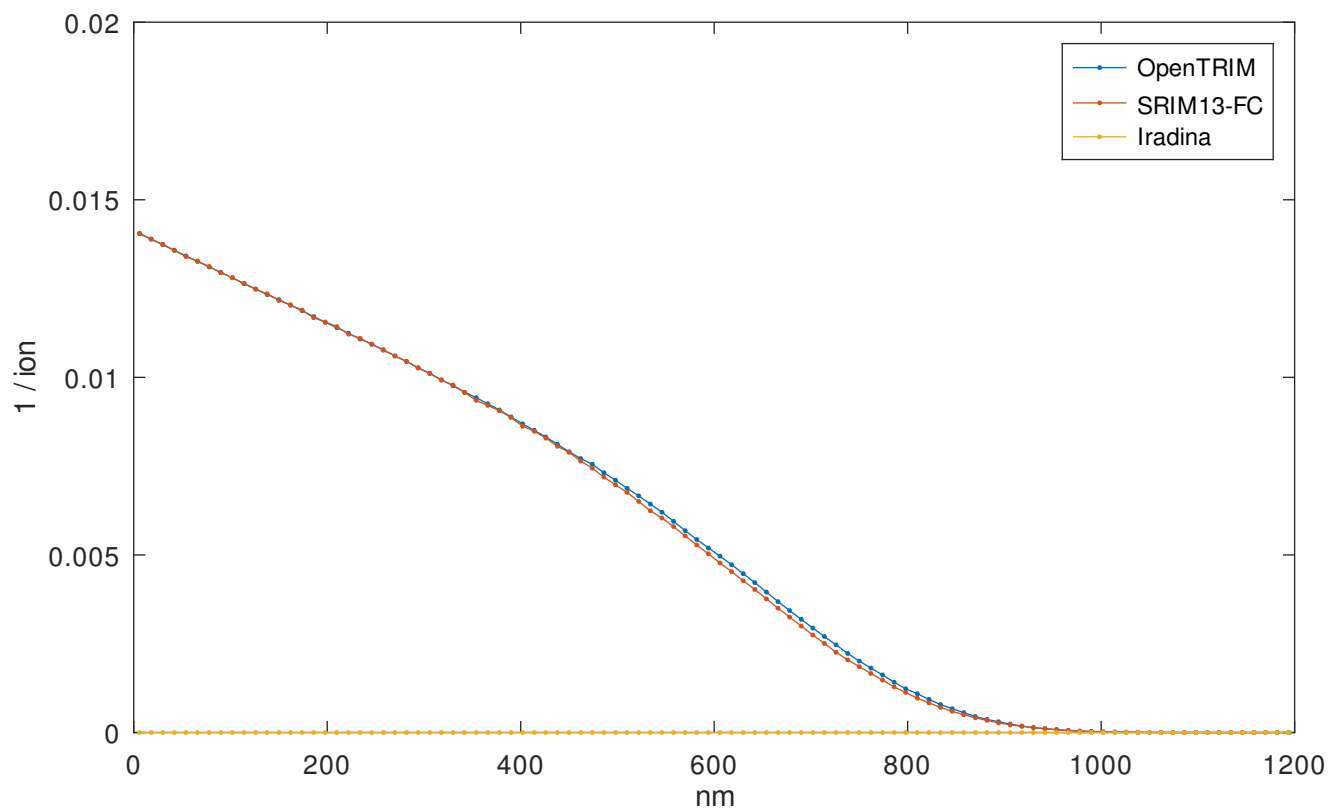
Replacements



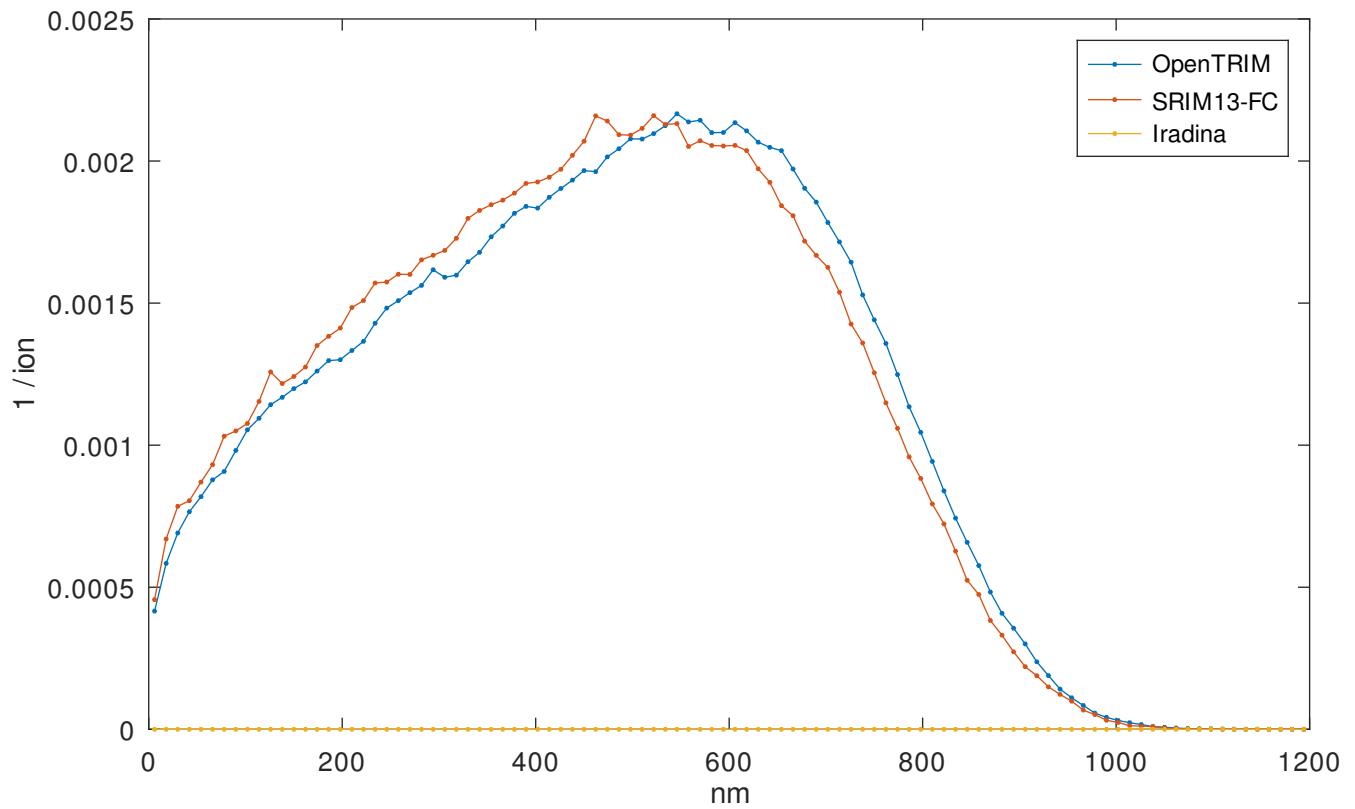
Implanted Fe ion



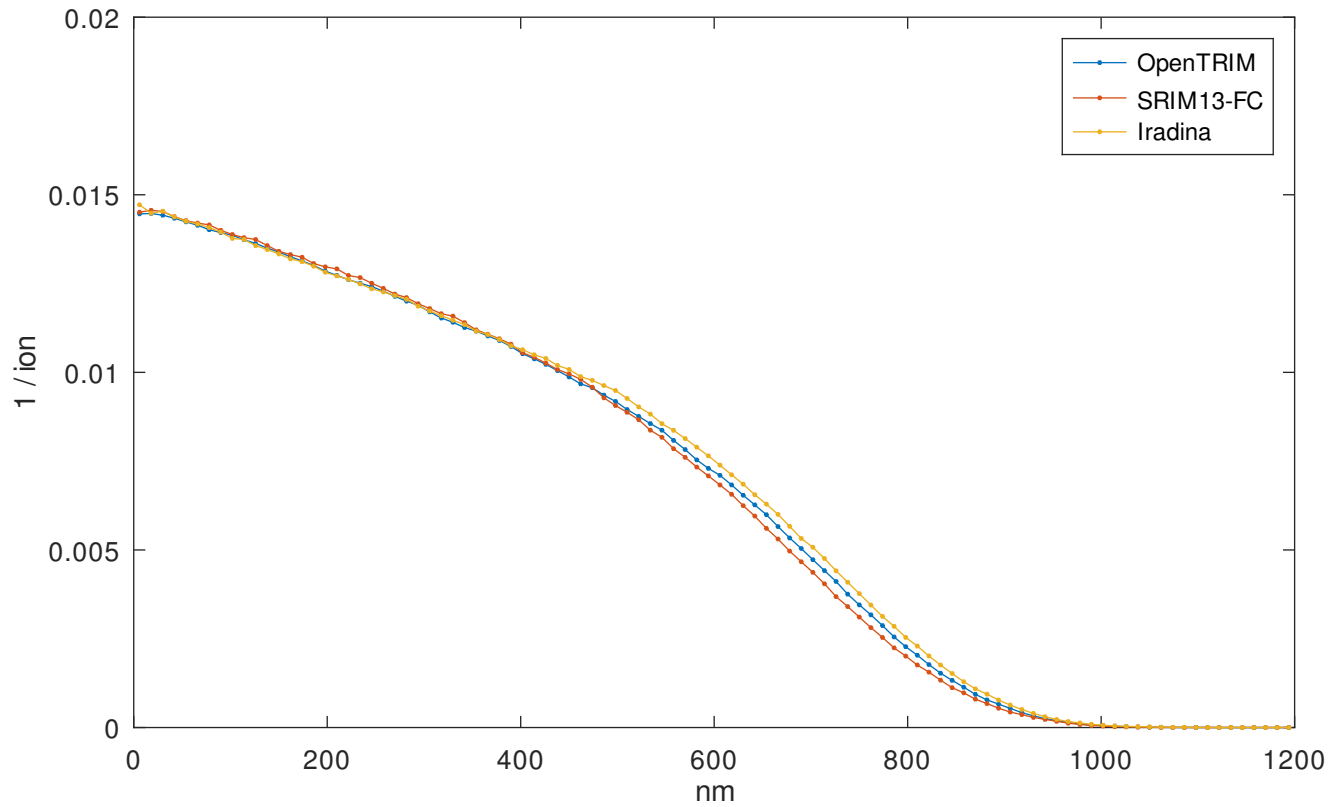
Electronic energy loss fraction $\Delta E_e/E_0$ by Fe ion



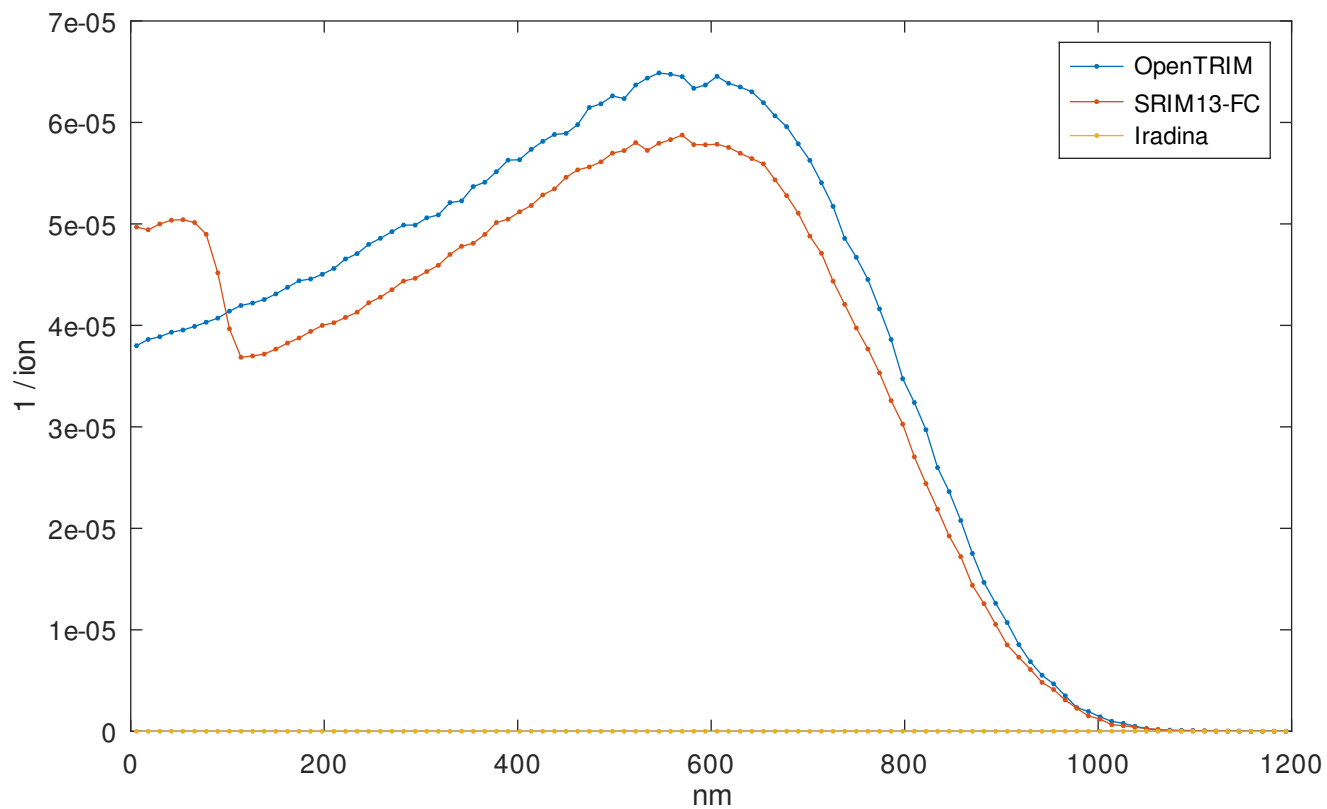
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



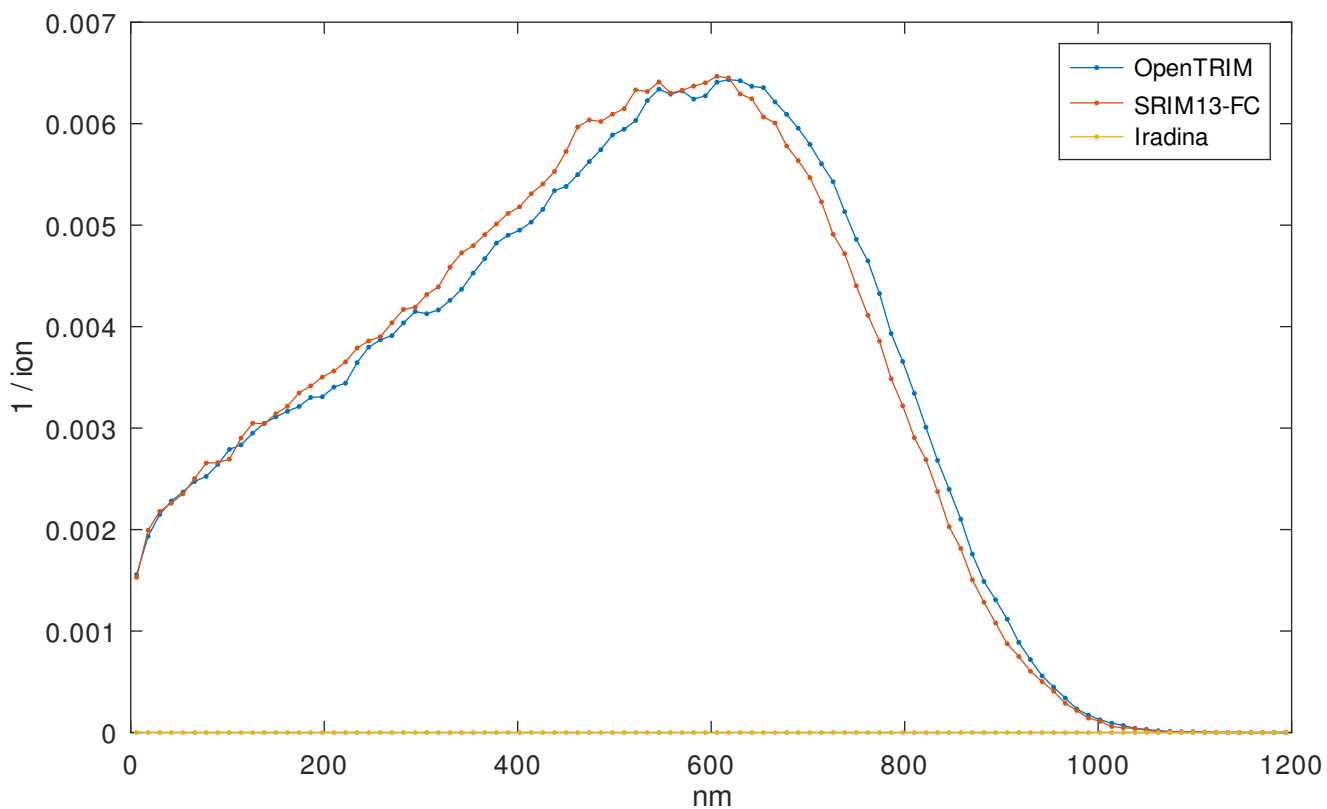
Total Electronic energy loss fraction $\Delta E_e/E_0$



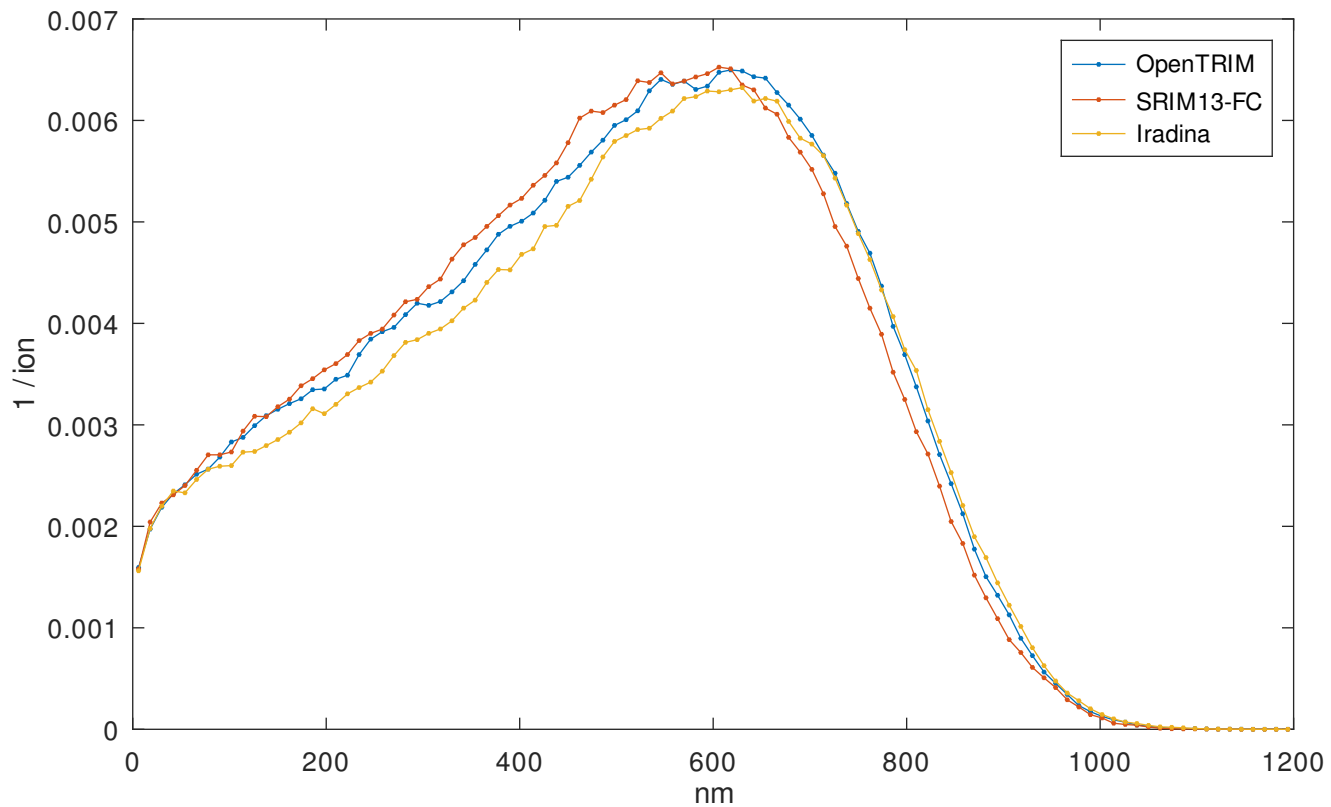
Nuclear energy loss fraction $\Delta E_n/E_0$ by Fe ion



Nuclear energy loss fraction $\Delta E_n/E_0$ by recoils



Total Nuclear energy loss fraction $\Delta E_n/E_0$



Total fractional energy loss $(\Delta E_e + \Delta E_n)/E_0$

